# STRAVA FITNESS DATA ANALYTICS

## EDA Insights Report

Prepared from the supplied Strava EDA notebook and case-study deck. This report focuses on the strongest, business-relevant findings rather than reproducing every chart.

## 1. Executive Summary

The analysis indicates that user engagement can be strengthened by encouraging consistent activity tracking, using activity relationships for personalized goals, and connecting fitness achievements to rewards or premium membership. The notebook specifically highlights daily steps, distance, calorie expenditure, activity logging behaviour and correlations among activity metrics.

## 2. Business Objectives

- Identify trends that can attract more customers and increase app usage.
- Understand behaviour patterns that can support subscription adoption.
- Analyze activity behaviour and patterns across key fitness metrics.
- Use customer activity choices and feedback signals to improve the app experience.

## 3. Data Preparation & Quality

The notebook combines multiple daily activity datasets, checks duplicates and missing values, and prepares a consolidated dataset for exploratory analysis. It reports 1,004 rows and 39 columns. A notable limitation is that some missing BMI, weight and sleep values were replaced with 0; the notebook itself recognizes that this can create unrealistic values.

## Daily Steps

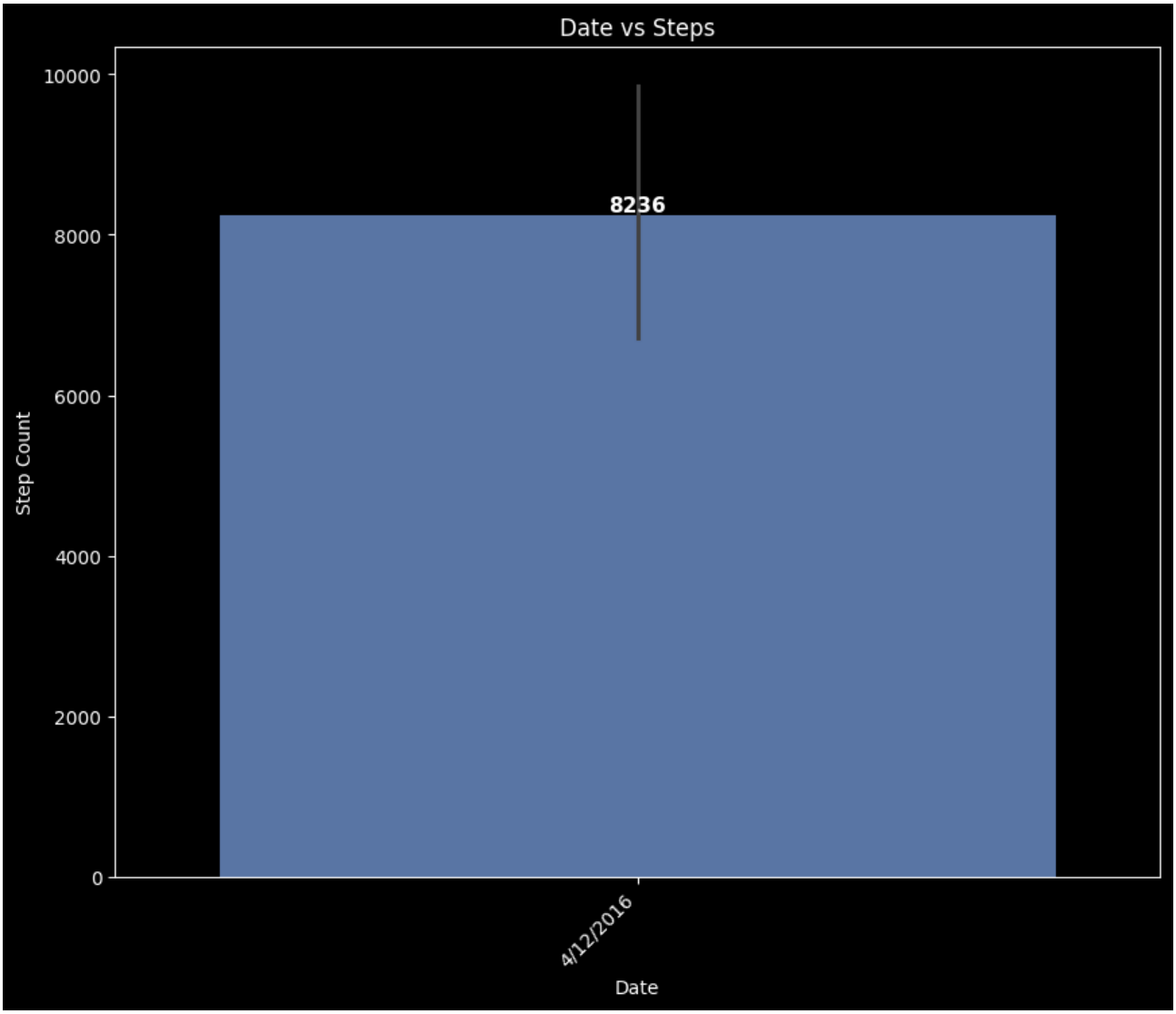



Figure 1. Daily steps visualization from the supplied notebook.

**Insight:** The notebook highlights recurring activity behaviour and gives an example daily total of 8,236 steps. Step activity is therefore a practical engagement metric for challenges, streaks and milestone notifications.

# Steps & Calories

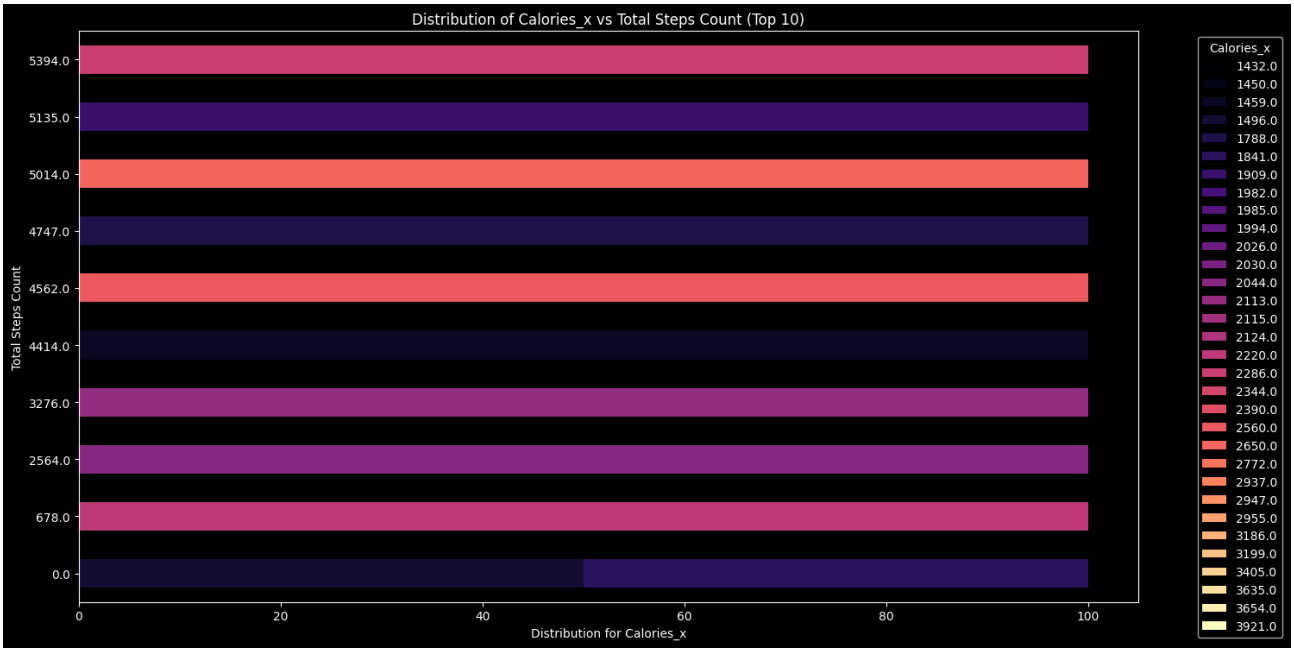



Figure 2. Steps and calorie expenditure visualization from the supplied notebook.

**Insight:** The notebook reports that calorie expenditure varies with step counts and that higher activity levels are generally associated with more calories burned. This supports personalized step and calorie goals.

# Tracked Distance vs Logged Activity

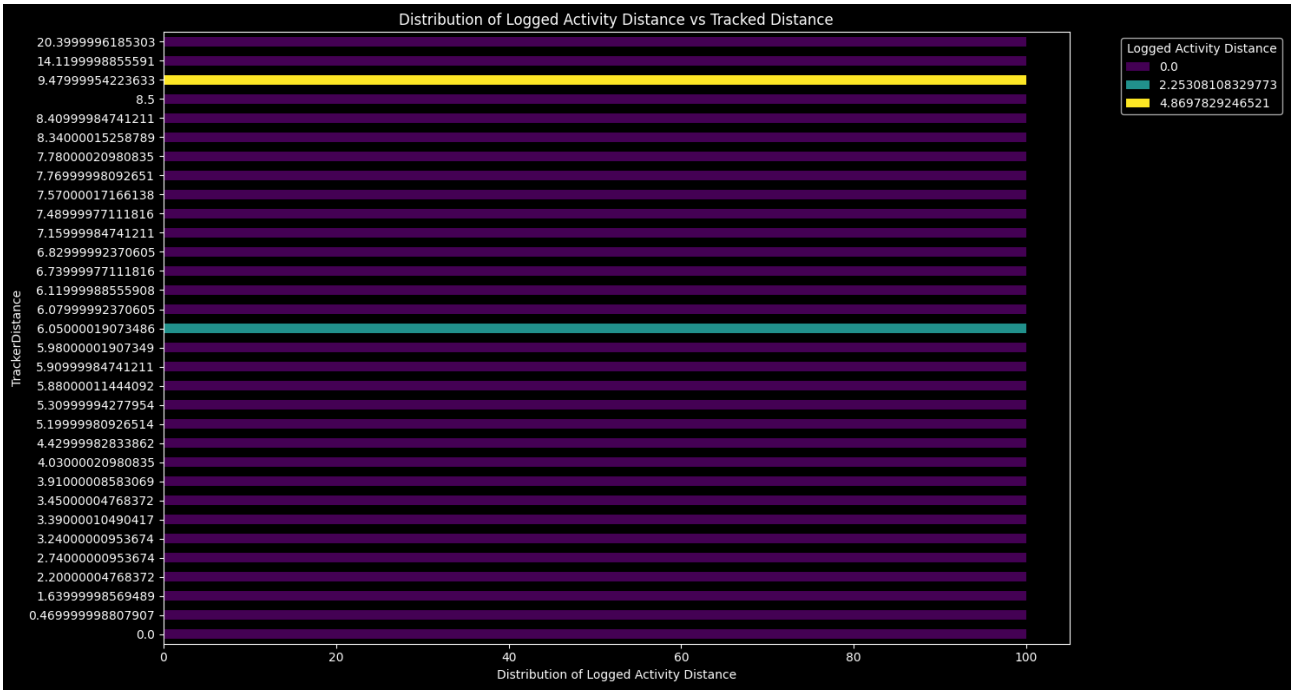



Figure 3. Tracked distance versus logged activity distribution from the supplied notebook.

**Insight:** Differences between tracked and logged activity suggest an opportunity to improve recording consistency and reduce friction in activity logging.

# Correlation Heatmap

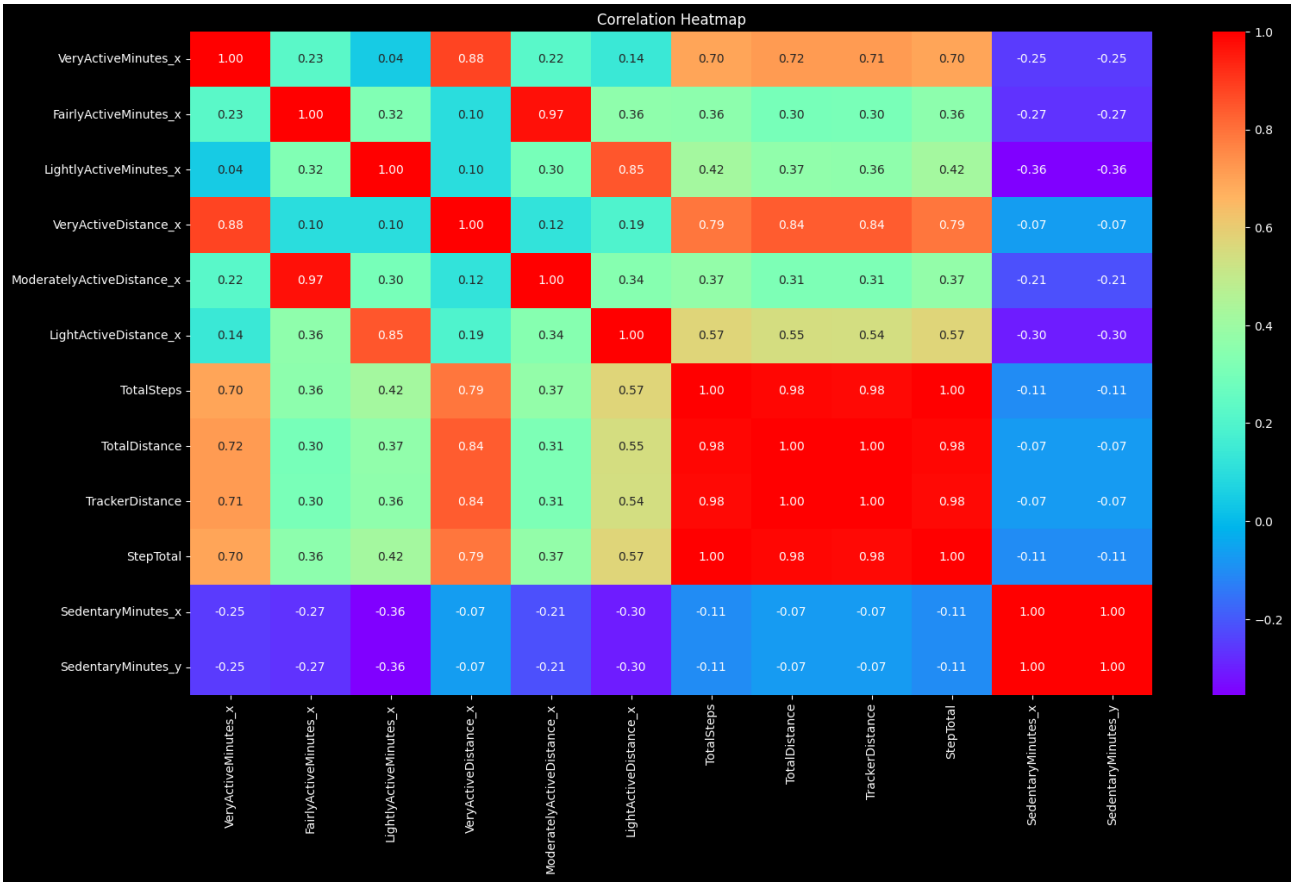



Figure 4. Correlation heatmap from the supplied notebook.

**Insight:** The notebook specifically identifies a strong positive relationship between steps and distance. These metrics can therefore work together for goal setting and progress tracking.

## 5. Business Recommendations

| Action | Recommendation |
|---|---|
| Improve activity logging | Encourage users to track and log daily distance alongside steps so they have more complete activity records. |
| Drive engagement with challenges | Use campaigns, challenges, webinars and reward programmes to motivate users and increase repeat activity. |
| Connect activity to premium value | Offer achievement-based benefits or discounts on paid memberships to encourage premium conversion. |
| Personalize goals | Use steps, distance and calorie relationships to create tailored targets and nudges. |
| Strengthen data quality | Review missing-value treatment before production deployment; avoid treating unknown physical measurements as genuine zero values. |

## 6. Conclusion

The strongest opportunity identified by the supplied EDA is to turn everyday activity behaviour into a repeat-engagement loop: track activity → provide personalized goals → motivate through challenges and rewards → connect achievements to premium value. Improving data-quality practices at the same time will make future personalization and business decisions more reliable.

## Source note

All analytical claims and selected visuals in this report are grounded in the supplied Strava_EDA.ipynb and its accompanying case-study presentation. The report deliberately uses a focused subset of charts rather than all notebook visualizations.